

Please check the examination details below before entering your candidate information

|                                                                                                      |                 |                                                                                                          |                                                                                                          |
|------------------------------------------------------------------------------------------------------|-----------------|----------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| Candidate surname                                                                                    |                 | Other names                                                                                              |                                                                                                          |
| <b>Pearson Edexcel</b><br><b>International</b><br><b>Advanced Level</b>                              |                 | Centre Number                                                                                            | Candidate Number                                                                                         |
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| Time 1 hour 30 minutes                                                                               | Paper reference | <b>WME01/01</b>                                                                                          |                                                                                                          |
| <b>Mathematics</b><br><b>International Advanced Subsidiary/Advanced Level</b><br><b>Mechanics M1</b> |                 |                                                                                                          |                                                                                                          |
| <b>You must have:</b><br>Mathematical Formulae and Statistical Tables (Yellow), calculator           |                 |                                                                                                          | Total Marks                                                                                              |

**Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.**

### Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided  
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Whenever a numerical value of  $g$  is required, take  $g = 9.8 \text{ m s}^{-2}$ , and give your answer to either 2 significant figures or 3 significant figures.

### Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 8 questions in this question paper. The total mark for this paper is 75.
- The marks for each question are shown in brackets  
– *use this as a guide as to how much time to spend on each question.*

### Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.
- If you change your mind about an answer, cross it out and put your new answer and any working underneath.
- Good luck with your examination.

Turn over ►



- (5)

Question 1 continued

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(Total 7 marks)

Q1





Question 2 continued

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Question 2 continued

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Q2





Question 3 continued

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### Question 3 continued



Question 3 continued

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Q3

(Total 9 marks)



4.

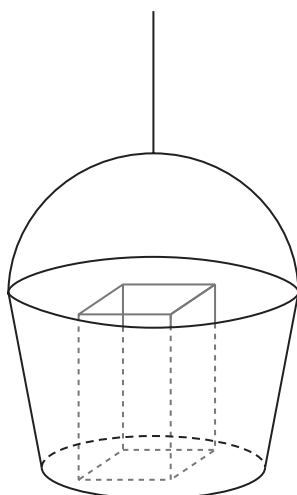
**Figure 1**

Figure 1 shows a large bucket used by a crane on a building site to move materials between the ground and the top of the building. The mass of the bucket is 15 kg.

The bucket is attached to a vertical cable with the bottom of the bucket horizontal. The cable is modelled as light and inextensible.

When the bucket is on the ground, a bag of cement of mass 25 kg is placed in the bucket.

The bucket with the bag of cement moves vertically upwards with constant acceleration  $0.2 \text{ m s}^{-2}$ . Air resistance is modelled as being negligible.

- (a) Find the tension in the cable. (3)

At the top of the building, the bag of cement is removed. A box of tools of mass 12 kg is now placed in the bucket.

Later on the bucket with the box of tools is moving vertically downwards with constant deceleration  $0.1 \text{ m s}^{-2}$ . Air resistance is again modelled as being negligible.

- (b) Find the magnitude of the normal reaction between the bucket and the box of tools. (3)

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Question 4 continued

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Question 4 continued

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Q4

(Total 6 marks)



5. [In this question  $\mathbf{i}$  and  $\mathbf{j}$  are perpendicular horizontal unit vectors.]

A particle  $P$  is moving with constant acceleration. At 2 pm, the velocity of  $P$  is  $(3\mathbf{i} + 5\mathbf{j}) \text{ km h}^{-1}$  and at 2.30 pm the velocity of  $P$  is  $(\mathbf{i} + 7\mathbf{j}) \text{ km h}^{-1}$

At time  $T$  hours after 2 pm,  $P$  is moving in the direction of the vector  $(-\mathbf{i} + 2\mathbf{j})$

(a) Find the value of  $T$ .

(6)

Another particle,  $Q$ , has velocity  $\mathbf{v}_Q \text{ km h}^{-1}$  at time  $t$  hours after 2 pm, where

$$\mathbf{v}_Q = (-4 - 2t)\mathbf{i} + (\mu + 3t)\mathbf{j}$$

and  $\mu$  is a constant.

Given that there is an instant when the velocity of  $P$  is equal to the velocity of  $Q$ ,

(b) find the value of  $\mu$ .

(3)

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**Question 5 continued**

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Question 5 continued

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Q5

(Total 9 marks)



- (c) Find the distance  $BC$ . (5)

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Question 6 continued

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Question 6 continued

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Lined area for writing the answer to Question 6.

(Total 13 marks)

Q6





Question 7 continued

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Question 7 continued

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Lined area for writing the answer to Question 7.

(Total 10 marks)

Q7



- (5)

Question 8 continued

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Question 8 continued

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Question 8 continued

Lined area for writing the answer to Question 8.

Q8

Grading box for Question 8.

(Total 13 marks)

TOTAL FOR PAPER: 75 MARKS

END

